

LESSON 1

Functions

Exercise 1.1

1. Tick (✓) the correct option for the given questions:

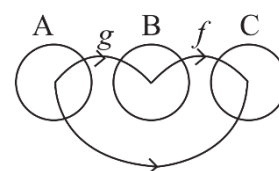
(A) Which of the following statements is incorrect regarding the composite functions?

- a) If $f: A \rightarrow B$ and $g: B \rightarrow C$ exist, and $A \rightarrow C$ is also a function, it is composite.
- b) If $g: A \rightarrow B$ and $f: B \rightarrow C$ are one-to-one and onto, then $A \rightarrow C$ is composite.
- c) If $g: A \rightarrow B$ and $f: B \rightarrow C$ are defined, the function $A \rightarrow C$ is denoted by $g \circ f$.
- d) If $f: A \rightarrow B$ and $g: B \rightarrow C$ are defined, the relation $C \rightarrow A$ cannot be composite.

Ans: c) If $g: A \rightarrow B$ and $f: B \rightarrow C$ are defined, the function $A \rightarrow C$ is denoted by $g \circ f$.

(B) In the given mapping diagram, how is the function defined from A to C denoted?

- a) $fg(x)$
- b) $gf(x)$
- c) $ff(x)$
- d) $gg(x)$



Ans: a) $fg(x)$

(C) If $f(x) = \{(3, -1), (2, 0), (1, 1)\}$ and $g(x) = \{(-1, 4), (0, 5), (1, 6)\}$, what is $gf(x)$?

- a) $\{(4, 3), (2, 5), (1, 6)\}$
- b) $\{(3, 4), (2, 5), (2, 6)\}$
- c) $\{(3, 4), (1, 5), (2, 6)\}$
- d) $\{(3, 4), (2, 5), (1, 6)\}$

Ans: d) $\{(3, 4), (2, 5), (1, 6)\}$

Solution:

$$\begin{aligned} \text{Here, } gf(x) &= g(f(x)) \\ gf(3) &= g(f(3)) = g(-1) = 4 \\ gf(2) &= g(f(2)) = g(0) = 5 \\ gf(1) &= g(f(1)) = g(1) = 6 \\ \therefore gf(x) &= \{(3, 4), (2, 5), (1, 6)\} \end{aligned}$$

(D) If $f(x) = 3x - 1$ and $g(x) = x + 1$, what is $fg(x)$?

- a) $3x - 1$
- b) $3x + 1$
- c) $3x - 2$
- d) $3x + 2$

Ans: d) $3x + 2$

Solution:

$$fg(x) = f(g(x)) = f(x + 1) = 3(x + 1) - 1 = 3x + 3 - 1 = 3x + 2$$

(E) If $f(x) = 1 - 3x$ and $g(x) = 2x + 1$, what is $gf(\frac{1}{2})$?

- a) 3
- b) 1
- c) 2
- d) 0

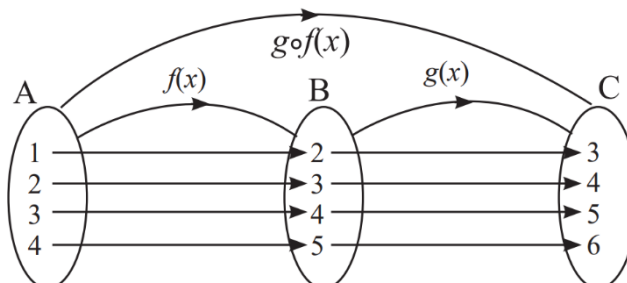
Ans: d) 0

Solution:

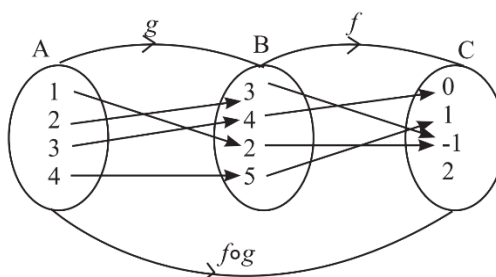
$$\begin{aligned} f(\frac{1}{2}) &= 1 - 3(\frac{1}{2}) = 1 - \frac{3}{2} = -\frac{1}{2} \\ gf(\frac{1}{2}) &= g(f(\frac{1}{2})) = g(-\frac{1}{2}) = 2(-\frac{1}{2}) + 1 = -1 + 1 = 0 \end{aligned}$$

2. Define composite function with a mapping diagram.

If functions $f: A \rightarrow B$ and $g: B \rightarrow C$ are defined, and a relation $A \rightarrow C$ is also defined as a function, then the function $A \rightarrow C$ is called the composite function of f and g . This is denoted in the form $g \circ f: A \rightarrow C$ or $g \circ f(x)$ or $g \circ f$ or $gf(x)$ or gf .



3. In the given mapping diagram, functions $g: A \rightarrow B$ and $f: B \rightarrow C$ are defined. Answer the following:



(a) Write function $g: A \rightarrow B$ as a set of ordered pairs.

Ans: Function $g: A \rightarrow B = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$

(b) Write function $f: B \rightarrow C$ as a set of ordered pairs.

Ans: Function $f: B \rightarrow C = \{(3, -1), (4, 0), (2, -1), (5, 1)\}$

(c) Find the values of $f \circ g(1)$, $f \circ g(2)$, $f \circ g(3)$, and $f \circ g(4)$.

Ans: $f \circ g(1) = f(g(1)) = f(2) = -1$

$f \circ g(2) = f(g(2)) = f(3) = -1$

$f \circ g(3) = f(g(3)) = f(4) = 0$

$f \circ g(4) = f(g(4)) = f(5) = 1$

(d) Write the composite function $f \circ g$ as a set of ordered pairs.

Ans: By combining the starting input from set A with the final resulting output in set C:

Composite function $f \circ g = \{(1, -1), (2, -1), (3, 0), (4, 1)\}$

4. If $f(x) = \{(1, 2), (2, 3), (3, 4)\}$ and $g(x) = \{(2, 4), (3, 5), (4, 6)\}$, show $gf(x)$ in a mapping diagram and write it as a set of ordered pairs.

Solution:

Given, $f(x) = \{(1, 2), (2, 3), (3, 4)\}$

$g(x) = \{(2, 4), (3, 5), (4, 6)\}$

then,

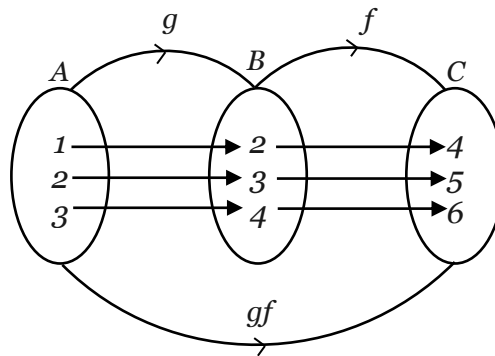
for $x=1$: $gf(1) = g(f(1)) = g(2) = 4$

for $x=2$: $gf(2) = g(f(2)) = g(3) = 5$

for $x=3$: $gf(3) = g(f(3)) = g(4) = 6$

Ans: $gf(x) = \{(1, 4), (2, 5), (3, 6)\}$

Mapping Diagram :



5. If $f(x) = 3x + 2$ and $g(x) = 5 - 3x$, find the value of the following composite functions:

a. $gf(x)$

Solution:

Given, $f(x) = 3x + 2$

$g(x) = 5 - 3x$

then,

$gf(x) = g(f(x)) = g(3x + 2)$

$= 5 - 3(3x + 2)$

$= 5 - 9x - 6$

$= -9x - 1$

Ans: $gf(x) = -9x - 1$

b. $fg(x)$

Solution:

$fg(x) = f(g(x)) = f(5 - 3x)$

$= 3(5 - 3x) + 2$

$= 15 - 9x + 2$

$= 17 - 9x$

Ans: $fg(x) = 17 - 9x$

c. $ff(x)$

Solution:

$$\begin{aligned}ff(x) &= f(f(x)) = f(3x + 2) \\&= 3(3x + 2) + 2 \\&= 9x + 6 + 2 \\&= 9x + 8\end{aligned}$$

Ans: $ff(x) = 9x + 8$

d. $ff(3)$

Solution:

$$\begin{aligned}ff(x) &= 9x + 8 \quad (\text{from question c}) \\ff(3) &= 9(3) + 8 = 27 + 8 = 35\end{aligned}$$

Ans: $ff(3) = 35$

6. If $f(x) = \frac{2x-1}{x+3}$ ($x \neq -3$) and $g(x) = \frac{3x+2}{x-4}$ ($x \neq 4$), find the value of $fg(x)$.

Solution:

Here, Given,

$$f(x) = \frac{2x-1}{x+3}, \quad g(x) = \frac{3x+2}{x-4}$$

Now,

$$\begin{aligned}fg(x) &= f(g(x)) \\&= f\left(\frac{3x+2}{x-4}\right) \\&= \frac{2\left(\frac{3x+2}{x-4}\right)-1}{\left(\frac{3x+2}{x-4}\right)+3} \\&= \frac{\frac{2(3x+2)-1(x-4)}{(3x+2)+3(x-4)}}{\frac{x-4}{x-4}} \\&= \frac{2(3x+2)-(x-4)}{(3x+2)+3(x-4)} \\&= \frac{6x+4-x+4}{3x+2+3x-12} \\&= \frac{5x+8}{6x-10}\end{aligned}$$

Ans: $\therefore fg(x) = \frac{5x+8}{6x-10}$

7. If $f(x) = 2x + 5$ and $g(x) = \frac{x-5}{2}$, prove that $fg(x)$ is an identity function.

Solution:

Here, Given,

$$\begin{aligned}f(x) &= 2x + 5 \\g(x) &= \frac{x-5}{2}\end{aligned}$$

Now,

$$\begin{aligned}fg(x) &= f(g(x)) \\&= f\left(\frac{x-5}{2}\right) \\&= 2\left(\frac{x-5}{2}\right) + 5 \\&= (x-5) + 5 \\&= x - 5 + 5 \\&= x\end{aligned}$$

Since $fg(x) = x$, the output of the function is identical to its input.

Ans: $\therefore fg(x)$ is an identity function.

8. a. If $f(x) = \frac{x-3}{2}$ and $fg(x) = x$, find the value of $g(x)$.

Solution:

Here, Given,

$$f(x) = \frac{x-3}{2} \quad \text{and} \quad fg(x) = x$$

Now,

$$f(g(x)) = x$$

$$\text{Or, } \frac{g(x)-3}{2} = x$$

$$\text{Or, } g(x) - 3 = 2x$$

$$\text{Or, } g(x) = 2x + 3$$

Ans: $\therefore g(x) = 2x + 3$

8. b. If $g(x) = 4 - x$ and $gh(x) = 11 - 2x$, find the value of $h(x)$.

Solution:

Here,

$$g(x) = 4 - x \quad \text{and} \quad gh(x) = 11 - 2x$$

Now,

$$g(h(x)) = 11 - 2x$$

$$\text{Or, } 4 - h(x) = 11 - 2x$$

$$\text{Or, } 4 - 11 + 2x = h(x)$$

$$\text{Or, } 2x - 7 = h(x)$$

Ans: $\therefore h(x) = 2x - 7$

9. a. If $f(x) = \frac{6}{x-2}$ ($x \neq 2$), $g(x) = ax^2 - 1$ and $gf(5) = 7$, find the value of a.

Solution:

Here, Given,

$$f(x) = \frac{6}{x-2}$$

$$g(x) = ax^2 - 1$$

$$gf(5) = 7$$

$$a = ?$$

Now,

$$gf(5) = g(f(5))$$

$$\text{Or, } gf(5) = g\left(\frac{6}{5-2}\right)$$

$$\text{Or, } gf(5) = g\left(\frac{6}{3}\right) = g(2)$$

$$\text{Or, } gf(5) = g(2) = 7$$

now,

$$g(2) = 7$$

$$\text{Or, } a(2)^2 - 1 = 7$$

$$\text{Or, } 4a = 8$$

$$\text{Or, } a = 2$$

Ans: $\therefore a = 2$

9. b. If $f(x) = ax + 5$, $g(x) = 8x + 13$ and $gf(5) = 93$, find the value of a.

Solution:

Here,

$$f(x) = ax + 5$$

$$g(x) = 8x + 13$$

$$gf(5) = 93$$

Now,

$$gf(5) = g(5a + 5)$$

$$\text{Or, } 93 = g(5a + 5)$$

$$\text{Or, } 93 = 8(5a + 5) + 13$$

$$\text{Or, } 93 = 40a + 40 + 13$$

$$\text{Or, } 93 - 53 = 40a$$

$$\text{Or, } 40 = 40a$$

$$\text{Or, } a = 1$$

Ans: $\therefore a = 1$

10. If $f(x) = kx + 1$, $g(x) = 3x - 7$ and $gf(1) = 2$, find the value of k .

Solution:

Here,

$$f(x) = kx + 1$$

$$g(x) = 3x - 7$$

$$gf(1) = 2$$

$$k = ?$$

Now,

$$gf(1) = g(k + 1)$$

$$\text{Or, } 2 = g(k + 1)$$

$$\text{Or, } 2 = 3(k + 1) - 7$$

$$\text{Or, } 2 = 3k + 3 - 7$$

$$\text{Or, } 2 + 4 = 3k$$

$$\text{Or, } 6 = 3k$$

$$\text{Or, } k = 2$$

Ans: $\therefore k = 2$

11. If $f(x) = 3x - 2$, $fg(x) = 6x - 2$ and $gf(x) = 8$, find the value of x .

Solution:

Here,

$$f(x) = 3x - 2$$

$$fg(x) = 6x - 2$$

$$gf(x) = 8$$

$$x = ?$$

Now,

$$fg(x) = 6x - 2$$

$$\text{Or, } f(g(x)) = 6x - 2$$

$$\text{Or, } 3(g(x)) - 2 = 6x - 2$$

$$\text{Or, } 3 \cdot g(x) = 6x - 2 + 2$$

$$\text{Or, } 3 \cdot g(x) = 6x$$

$$\text{Or, } g(x) = \frac{6x}{3}$$

$$\text{Or, } g(x) = 2x$$

Now,

$$gf(x) = 8$$

$$\text{Or, } g(3x - 2) = 8$$

$$\text{Or, } 2(3x - 2) = 8$$

$$\text{Or, } 6x - 4 = 8$$

$$\text{Or, } 6x = 8 + 4$$

$$\text{Or, } x = \frac{12}{6}$$

$$\text{Or, } x = 2$$

Ans: $\therefore x = 2$